

D181

MSCIN Capstone

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A curriculum specialist who has presented at two state conferences asks: “Does using a new graphic-organizer routine increase ninth graders' essay scores?” Her district uses a four-day week. Which design fits the question?

- A. A descriptive survey of teacher opinions.
- B. A study of the four-day schedule.
- C. An experimental or quasi-experimental design comparing classes with and without the routine.
- D. A correlational study of organizer use and scores.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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2.

A candidate at a school with a strong arts program first surveys 150 students and finds that math anxiety dropped after a new unit. She then interviews eight students to learn why. The survey used a 5-point scale. Which mixed-methods design is this?

- A. Convergent parallel.
- B. Purely quantitative, since the survey came first.
- C. Exploratory sequential.
- D. Explanatory sequential.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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3. [TRANSFER — reversed stem / different variable]

A researcher needs to build a new survey about student belonging but has no good existing items. Which mixed-methods order fits?

- A. Exploratory sequential: interviews first to generate items, then a survey to test them.
- B. Explanatory sequential: survey first, then interviews.
- C. Survey only, using any belonging questions found online.
- D. Convergent: survey and interviews at the same time.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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4.

A teacher with 22 students in her own gifted class, all from one suburban school, finds a new writing unit raised scores. The school recently won a technology grant. What claim can she reasonably make?

- A. The technology grant caused the gains.
- B. The unit raises writing scores for all students in the state.
- C. The results apply to all gifted students, since 22 is a large class.
- D. The unit was associated with gains for her class; wider use needs further study.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A single-group study measured reading fluency in September and May for 30 second graders using a new phonics routine. The school library also added 2,000 books. Students' fluency rose sharply. The teacher's room faces the playground. Which is a threat to internal validity?

- A. History and maturation: the new books and normal growth could explain the gain.
- B. Low reliability, because 30 students is a small group.
- C. None, because the gain was large.
- D. The playground noise.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Students took the same 20-item test in October and again in November and scored higher the second time, with no teaching in between. Which threat best explains the gain?

- A. The intervention.
- B. Testing: practice with the same items.
- C. History.
- D. Maturation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A candidate uses a vocabulary quiz with a clean, modern layout to measure reading comprehension. Students get nearly the same scores when retested a week later. What is true of the quiz?

- A. It is valid because scores are stable.
- B. It is neither reliable nor valid.
- C. It is reliable but not a valid measure of comprehension.
- D. It is valid because the layout is clear.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

A rubric targets exactly the writing skill in the standard, but two trained raters scoring the same essays often give different scores. What is the main problem?

- A. No problem, since the rubric matches the standard.
- B. Low reliability (inter-rater consistency).
- C. Sampling error.
- D. Low validity only; reliability is fine.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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9.

In a study run during a snowy winter term, classes using a new math routine averaged 78 on the unit test, while comparison classes averaged 72. The standard deviation was 12. The p-value was 0.03. What is the effect size (Cohen's d)?

- A. 2.0
- B. 6, a large effect.
- C. 0.03
- D. 0.5, a medium effect.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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10. [TRANSFER — reversed stem / different variable]

A study reports $d = 0.4$. The comparison group averaged 70 and the standard deviation was 10. What was the treatment group's mean?

- A.** 66
- B.** 74
- C.** 80
- D.** 70.4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A candidate has transcripts from 10 teacher interviews, recorded in a quiet library room, about a new science curriculum. What is the sound next step in analysis?

- A.** Code segments of each transcript, group related codes, and develop themes supported across interviews.
- B.** Average the interviews into one score.
- C.** Count how often the word “science” appears.
- D.** Pick the three most striking quotes that support the curriculum.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A candidate at a school with a new makerspace asks: “Does project-based learning improve engagement?” Her advisor says the study cannot be repeated as written. What should she add?

- A.** A longer literature review.
- B.** A larger sample.
- C.** An operational definition of engagement, such as percent of students on task in timed observations.
- D.** A description of the makerspace.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A curriculum specialist ~~who has presented at two state conferences~~ asks: "Does using a new graphic-organizer routine increase ninth graders' essay scores?" ~~Her district uses a four-day week.~~ Which design fits the question?

Target word: design for a causal question **Decides:** "Does X increase Y" is causal.

A	Confusion set	Describes views; does not test effect.
B	Camouflage	The schedule is not the variable in the question.
C	Correct	Causal questions need a comparison.
D	Adjacent category	Shows relationship, not cause.

2. Answer D Concept 2

A candidate ~~at a school with a strong arts program~~ first surveys 150 students and finds that math anxiety dropped after a new unit. She then interviews eight students to learn why. ~~The survey used a 5-point scale.~~ Which mixed-methods design is this?

Target word: name the design **Decides:** Numbers first, then qualitative to explain.

A	Confusion set	Convergent collects both strands at the same time.
B	Dropped element	Leaves out the interview strand.
C	Inverted operation	That design starts with qualitative work.
D	Correct	Quantitative first; qualitative explains the results.

3. Answer A Concept 2 • transfer

A researcher needs to build a new survey about student belonging but has no good existing items. Which mixed-methods order fits?

Target word: choose design from the purpose **Decides:** Qualitative first to build the tool.

A	Correct	Qualitative exploration builds the instrument.
B	Inverted operation	There is no survey to start with.
C	Adjacent category	Skips the need to build a fitting tool.
D	Confusion set	The survey cannot run before its items exist.

4. Answer D Concept 3

A teacher with 22 students in her own gifted class, all from one suburban school, finds a new writing unit raised scores. ~~The school recently won a technology grant.~~ What claim can she reasonably make?

Target word: limit of generalization **Decides:** Convenience sample of one gifted class.

A	Camouflage	The grant was not studied.
B	Absolute language	Overreaches from one class.
C	Wrong stem number	Class size does not make a sample representative.
D	Correct	Findings stay close to the sample studied.

5. Answer A Concept 4

A single-group study measured reading fluency in September and May for 30 second graders using a new phonics routine. The school library also added 2,000 books. Students' fluency rose sharply. ~~The teacher's room faces the playground.~~ Which is a threat to internal validity?

Target word: rival explanation **Decides:** Outside events and a school year of growth.

A	Correct	Both are rival explanations in a single-group design.
B	Confusion set	Reliability is about consistency of the measure, not group size.
C	Absolute language	Size of gain does not rule out rival causes.
D	Camouflage	Unrelated to the gain.

6. Answer B Concept 4 • transfer

Students took the same 20-item test in October and again in November and scored higher the second time, with no teaching in between. Which threat best explains the gain?

Target word: name the threat from the result **Decides:** Same test repeated with no teaching.

A	Dropped element	There was no intervention.
B	Correct	Familiarity with the pretest raises posttest scores.
C	Adjacent category	No outside event is described.
D	Confusion set	One month of growth is unlikely without teaching.

7. Answer C Concept 5

A candidate uses a vocabulary quiz ~~with a clean, modern layout~~ to measure reading comprehension. Students get nearly the same scores when retested a week later. What is true of the quiz?

Target word: reliable vs. valid **Decides:** Stable scores; wrong construct.

A	Inverted operation	Stability is reliability, not validity.
B	Dropped element	Ignores the stable retest results.
C	Correct	Consistent, but it measures vocabulary, not comprehension.
D	Camouflage	Layout does not show what is measured.

8. Answer B Concept 5 • transfer

A rubric targets exactly the writing skill in the standard, but two trained raters scoring the same essays often give different scores. What is the main problem?

Target word: diagnose — reversed case: valid target, inconsistent scores **Decides:** Scores are not consistent across raters.

A	Dropped element	Ignores rater disagreement.
B	Correct	Inconsistent scoring is a reliability problem, which also limits validity.
C	Adjacent category	Sampling concerns who is studied, not scorer agreement.
D	Inverted operation	The target is right; consistency is what fails.

9. Answer D Concept 6

In a study ~~run during a snowy winter term~~, classes using a new math routine averaged 78 on the unit test, while comparison classes averaged 72. The standard deviation was 12. ~~The p-value was 0.03.~~ What is the effect size (Cohen's d)?

Target word: compute d **Decides:** $(78 - 72) / 12$.

A	Inverted operation	Divides 12 by 6.
B	Dropped element	The raw difference without dividing by the standard deviation.
C	Camouflage	The p-value, not the effect size.
D	Correct	$6 / 12 = 0.5$.

10. Answer B Concept 6 · transfer

A study reports $d = 0.4$. The comparison group averaged 70 and the standard deviation was 10. What was the treatment group's mean?

Target word: solve for the treatment mean **Decides:** Difference = $0.4 \times 10 = 4$.

A	Inverted operation	Subtracts the difference.
B	Correct	$70 + 4 = 74$.
C	Wrong stem number	Adds the standard deviation itself.
D	Dropped element	Adds d without multiplying by the standard deviation.

11. Answer A Concept 7

A candidate has transcripts from 10 teacher interviews, ~~recorded in a quiet library room~~, about a new science curriculum. What is the sound next step in analysis?

Target word: qualitative analysis steps **Decides:** Code, group, and build themes.

A	Correct	Systematic coding builds evidence-based themes.
B	Inverted operation	Applies quantitative steps to qualitative data.
C	Confusion set	Word counts do not capture meaning.
D	Adjacent category	Cherry-picking, not analysis.

12. Answer C Concept 8

A candidate ~~at a school with a new makerspace~~ asks: "Does project-based learning improve engagement?" Her advisor says the study cannot be repeated as written. What should she add?

Target word: fix an undefined variable **Decides:** "Engagement" has no stated measure.

A	Adjacent category	Does not define the measure.
B	Dropped element	Sample size does not fix an undefined variable.
C	Correct	Defines the variable by how it is measured.
D	Camouflage	The setting is not the variable.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Question type sets the design	Descriptive questions (what is) call for surveys or observation; relational questions (is X related to Y) call for correlational designs; causal questions (does X change Y) call for experimental or quasi-experimental designs.	Using a correlational design to answer a does-it-cause question.	The verb in the research question: describe, relate, or affect.
2	Mixed-methods sequence	Explanatory sequential collects numbers first, then interviews to explain them; exploratory sequential starts with qualitative work to build a tool or ideas, then tests them with numbers; convergent collects both at once and compares.	Mixing up which strand comes first.	The order and purpose of the qualitative and quantitative phases.
3	Sampling and generalizability	Results generalize only as far as the sample represents the population; a convenience sample of one's own class limits how far findings apply.	Claiming findings from one class apply to all students in the state.	Who was studied vs. who the claim is about.
4	Threats to internal validity	History (outside events), maturation (natural growth), and testing (practice from a pretest) can explain changes in a single-group pretest-posttest study.	Crediting the intervention for any gain in a single-group study.	A gain plus an event or passage of time during the study.
5	Reliability vs. validity	Reliability is consistency of a measure; validity is whether it measures what it claims to. A measure can be reliable but not valid.	Assuming consistent scores mean the tool measures the right thing.	An instrument that gives stable results but targets the wrong skill, or vice versa.
6	Effect size	Cohen's $d = (\text{treatment mean} - \text{comparison mean}) / \text{standard deviation}$; about 0.2 is small, 0.5 medium, 0.8 large, and it shows practical size apart from statistical significance.	Reporting only the p-value, or dividing by the wrong number.	Two group means and a standard deviation.
7	Qualitative coding to themes	Qualitative analysis labels segments of data with codes, groups related codes into categories, and builds themes supported by evidence across participants.	Picking vivid quotes that fit the hoped-for answer, or counting words as analysis.	Interview or open-response data and a question about analysis steps.

#	Concept	Rule	Trap	Tell
8	Operational definitions	Define each variable by how it will be measured (for example, “engagement = minutes on task recorded every 5 minutes”) so others can repeat the study.	Leaving key terms like engagement or success undefined.	A variable named in a question with no stated measure.

Course scope (public description): The Master of Science in Curriculum and Instruction Capstone is the culminating course of the degree. It unites content area knowledge with the completion of a research project or study. This course teaches candidates, under the guidance of program faculty, to apply their data literacy and research skills to topics related to curriculum and instruction and to their career goals. Projects for this course include action research or applied research through the necessary qualitative, quantitative, or mixed research methods. Prerequisites for this course include Data- Informed Practices and Educational Research, as well as all prescribed courses in the candidates' area of study. Additionally, students wishing to add the Capstone with fewer than four weeks remaining in the term must receive permission from program leadership.